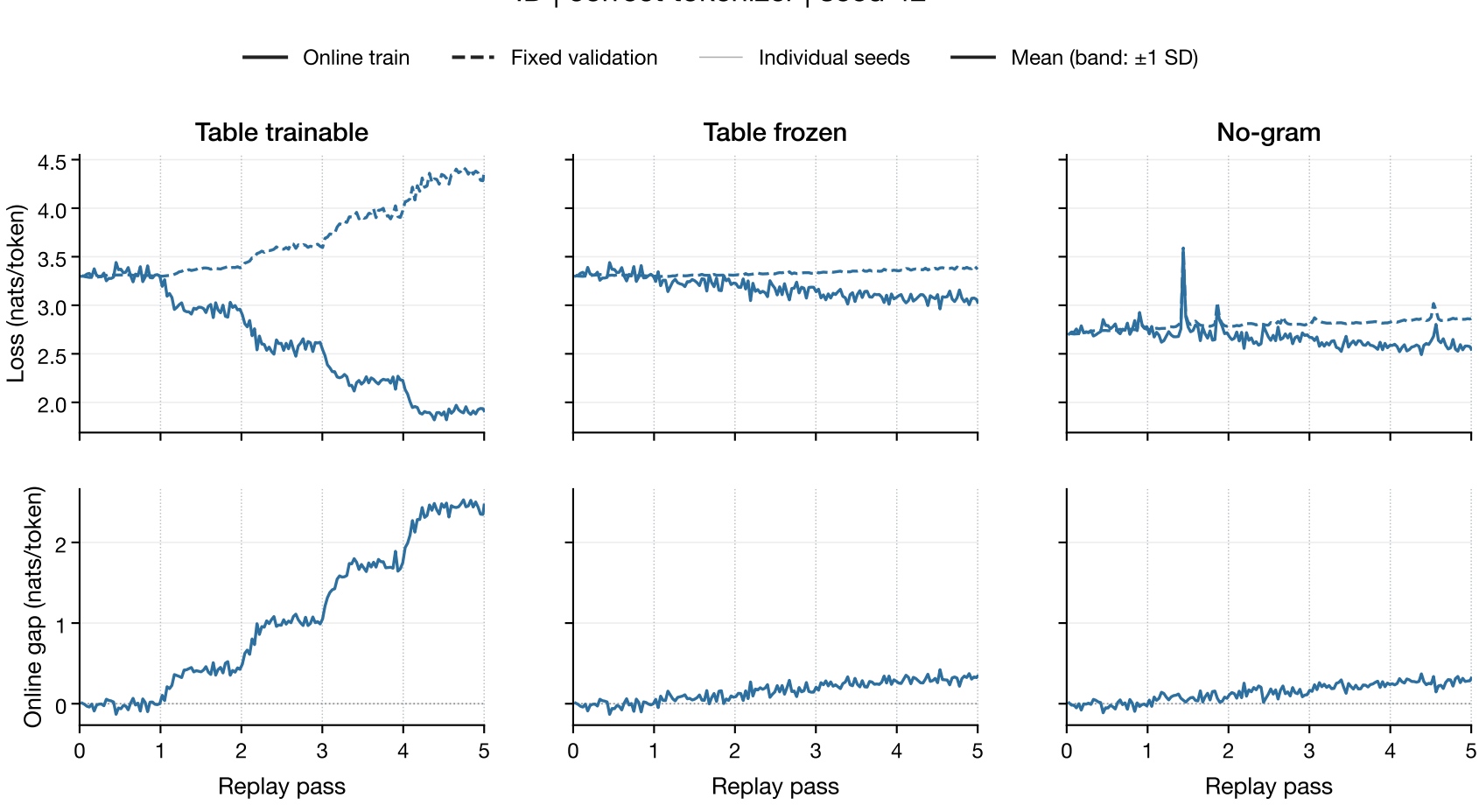
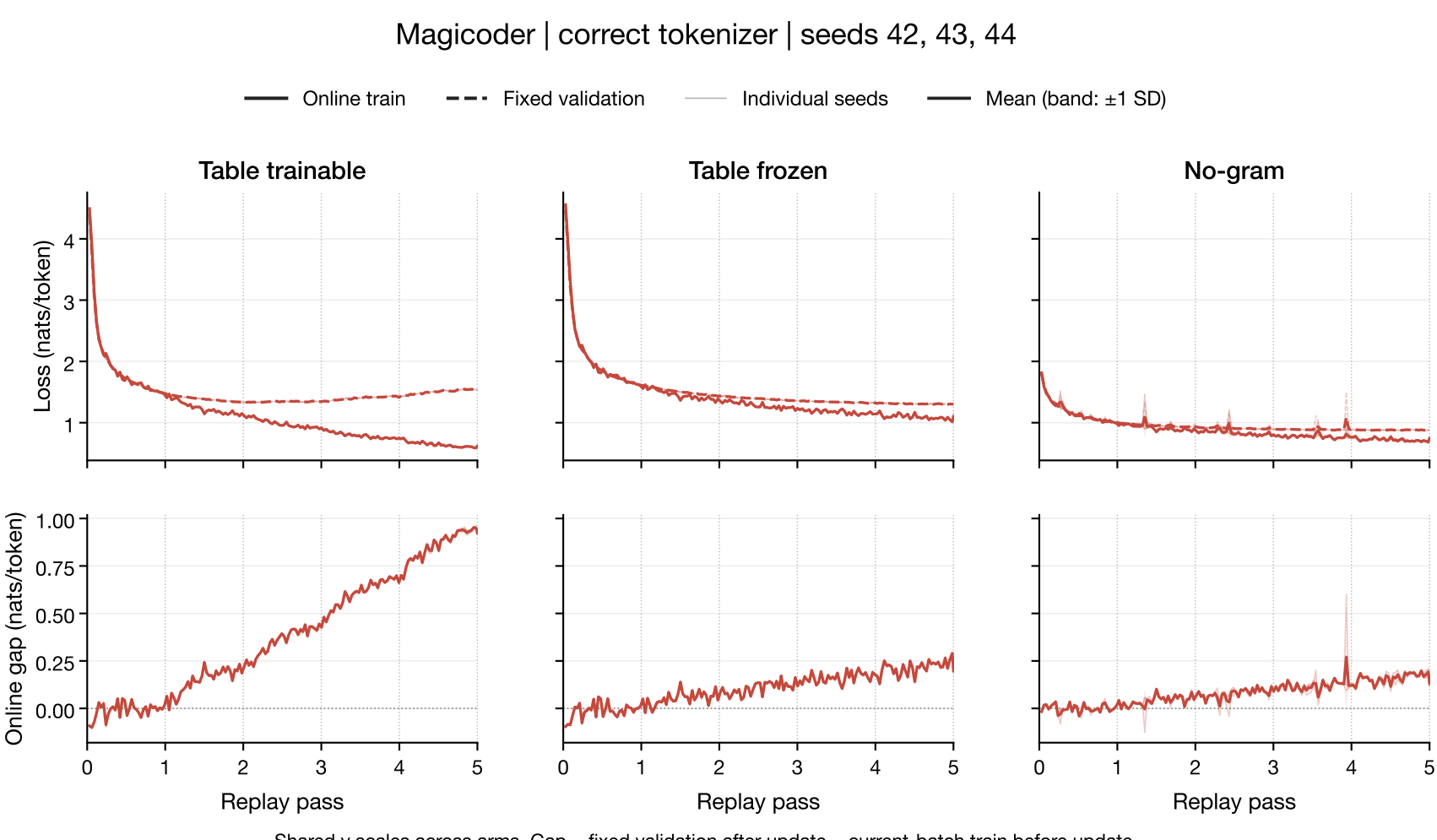
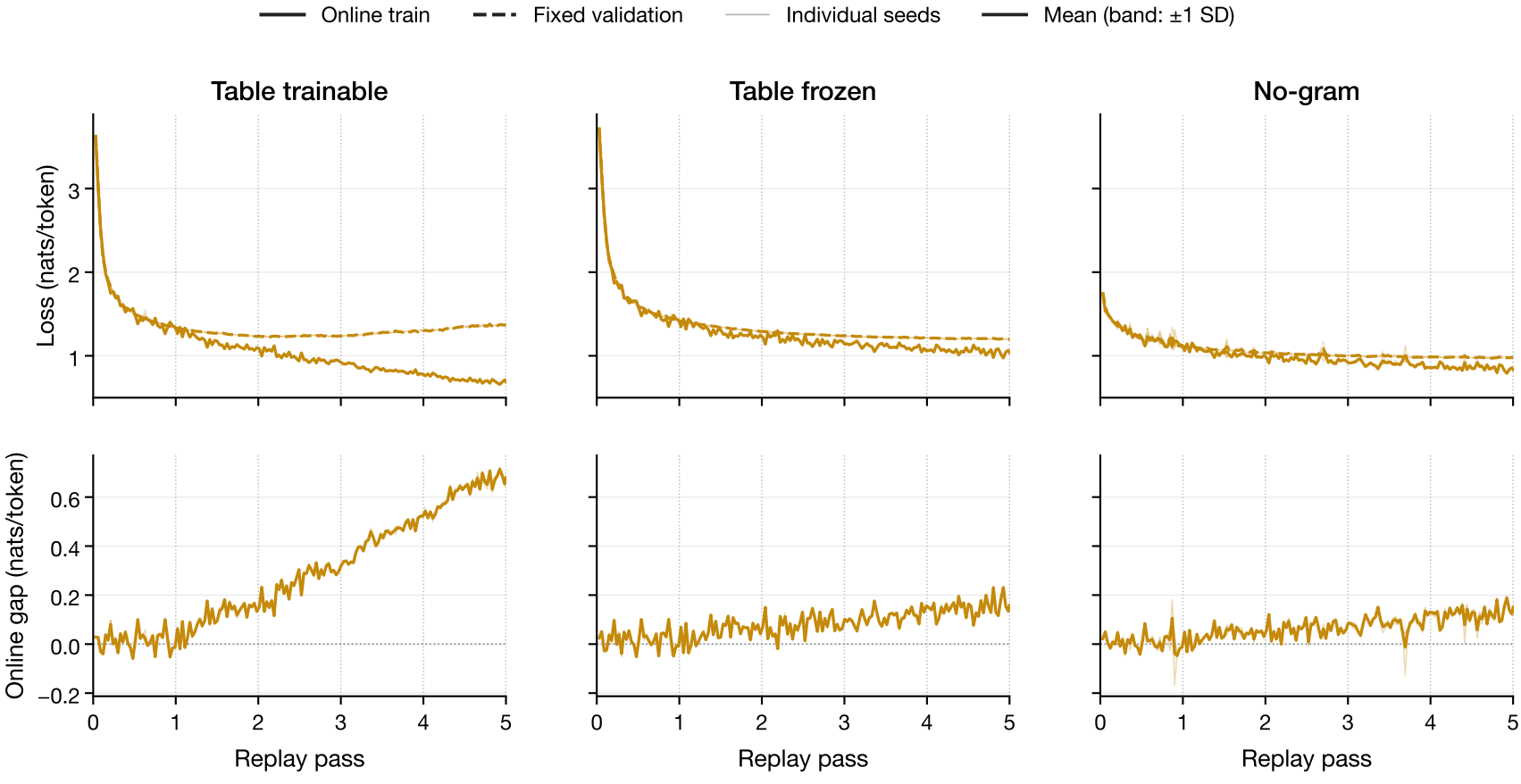
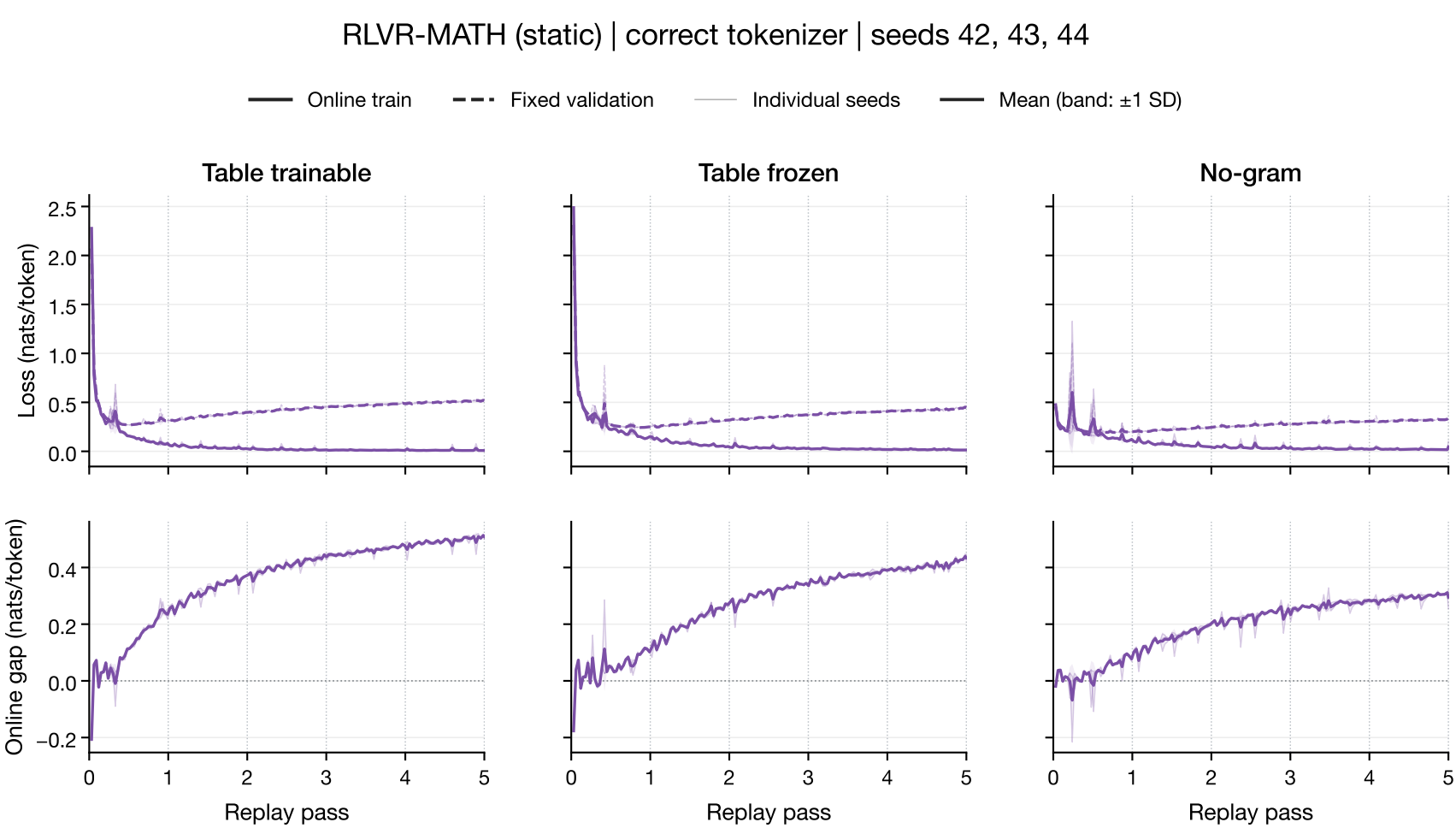
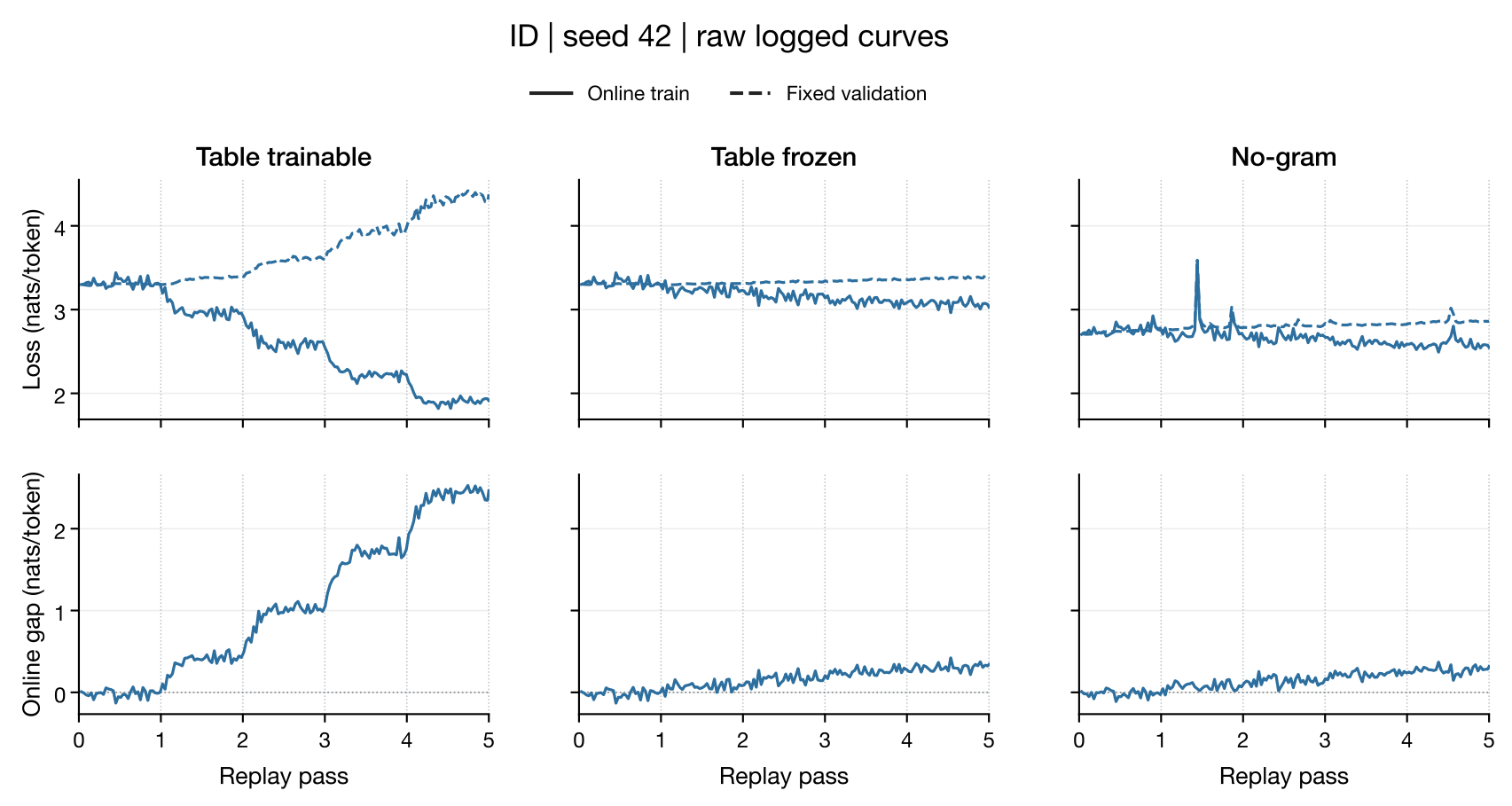
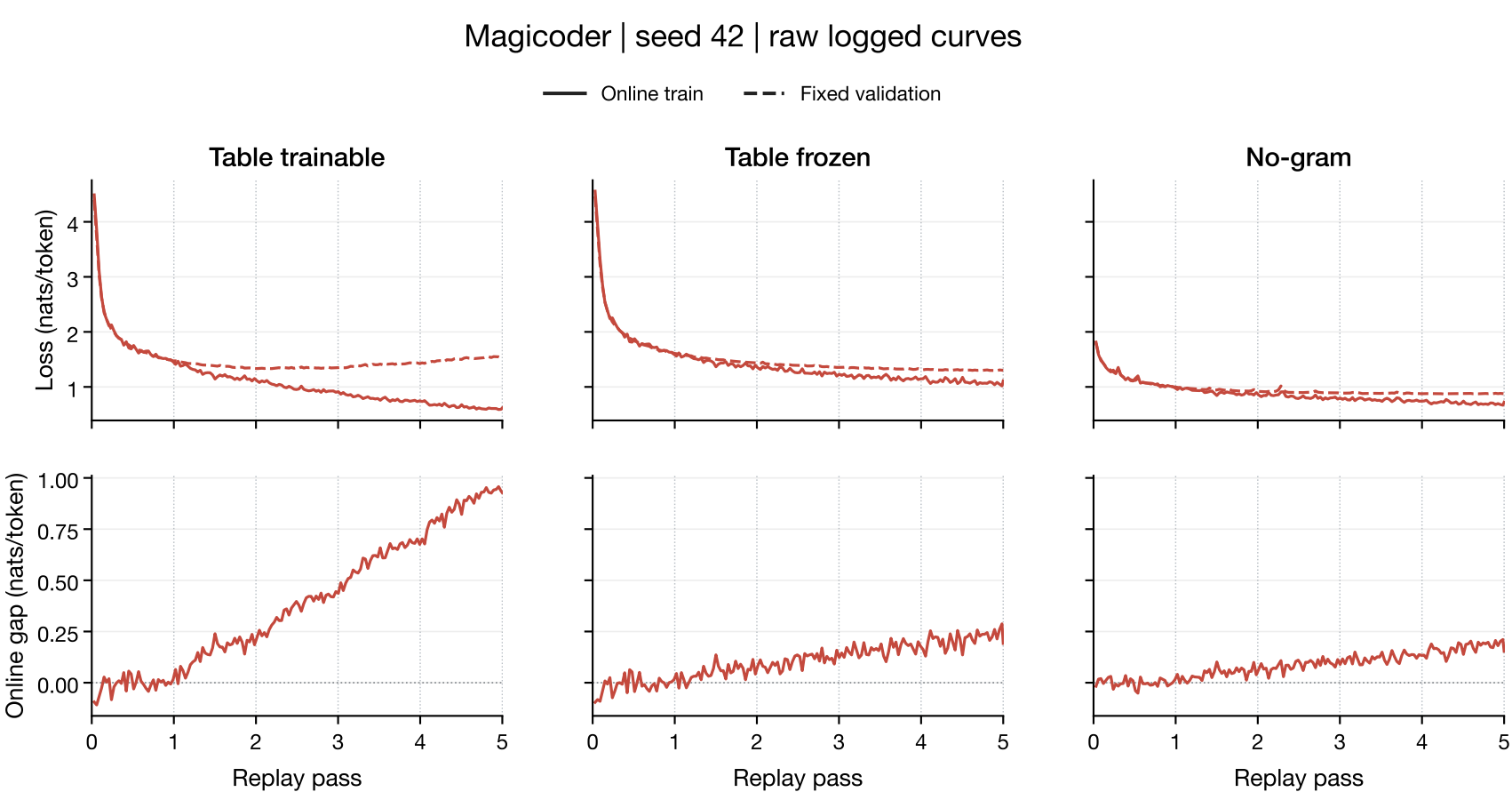

ID | correct tokenizer | seed 42
Online train
Fixed validation
Individual seeds
Mean (band: ±1 SD)
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Shared y scales across arms. Gap = fixed validation after update − current-batch train before update.

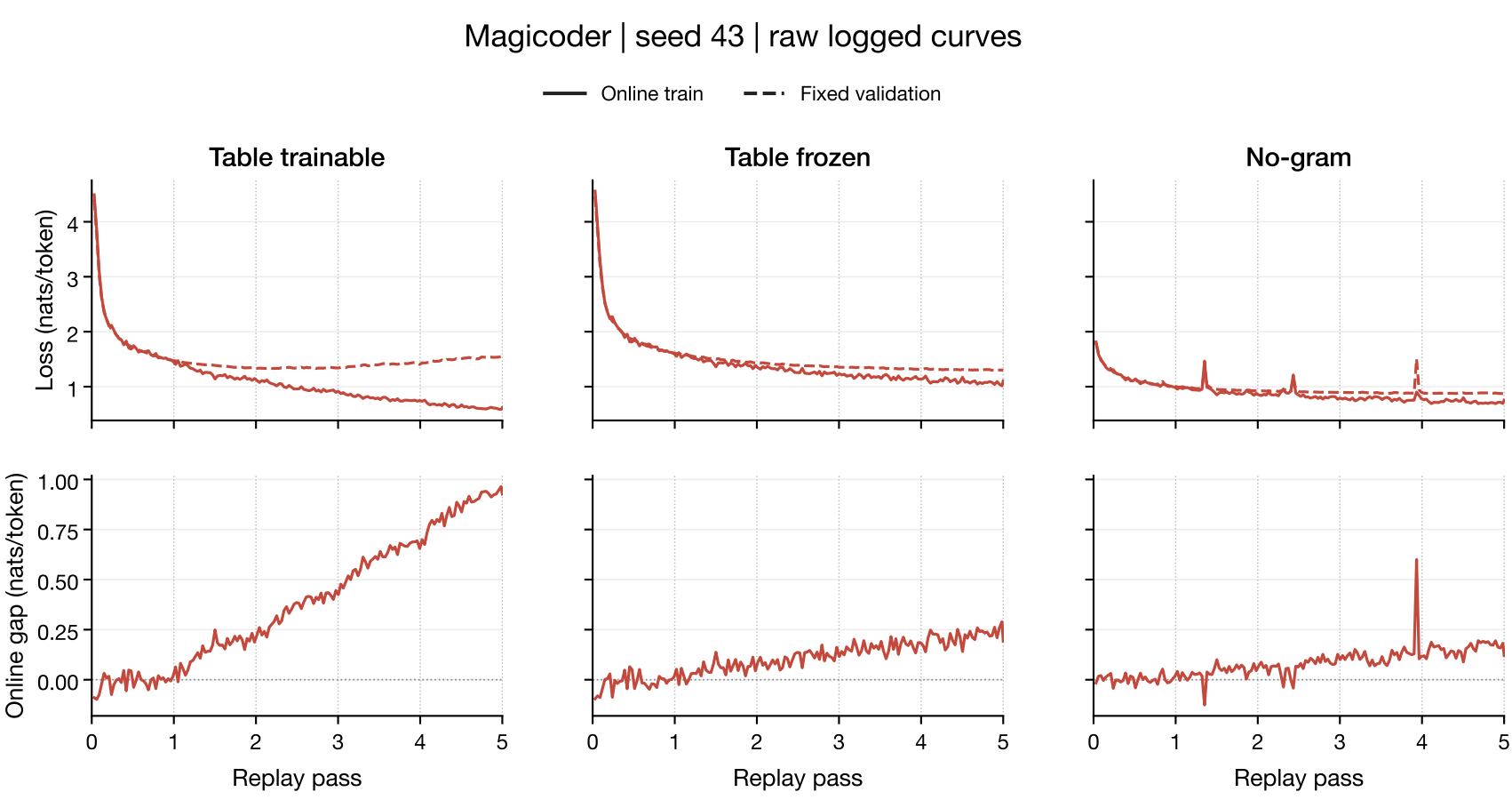

Magicoder | correct tokenizer | seeds 42, 43, 44
Online train
Fixed validation
Individual seeds
Mean (band: ±1 SD)
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Shared y scales across arms. Gap = fixed validation after update − current-batch train before update.

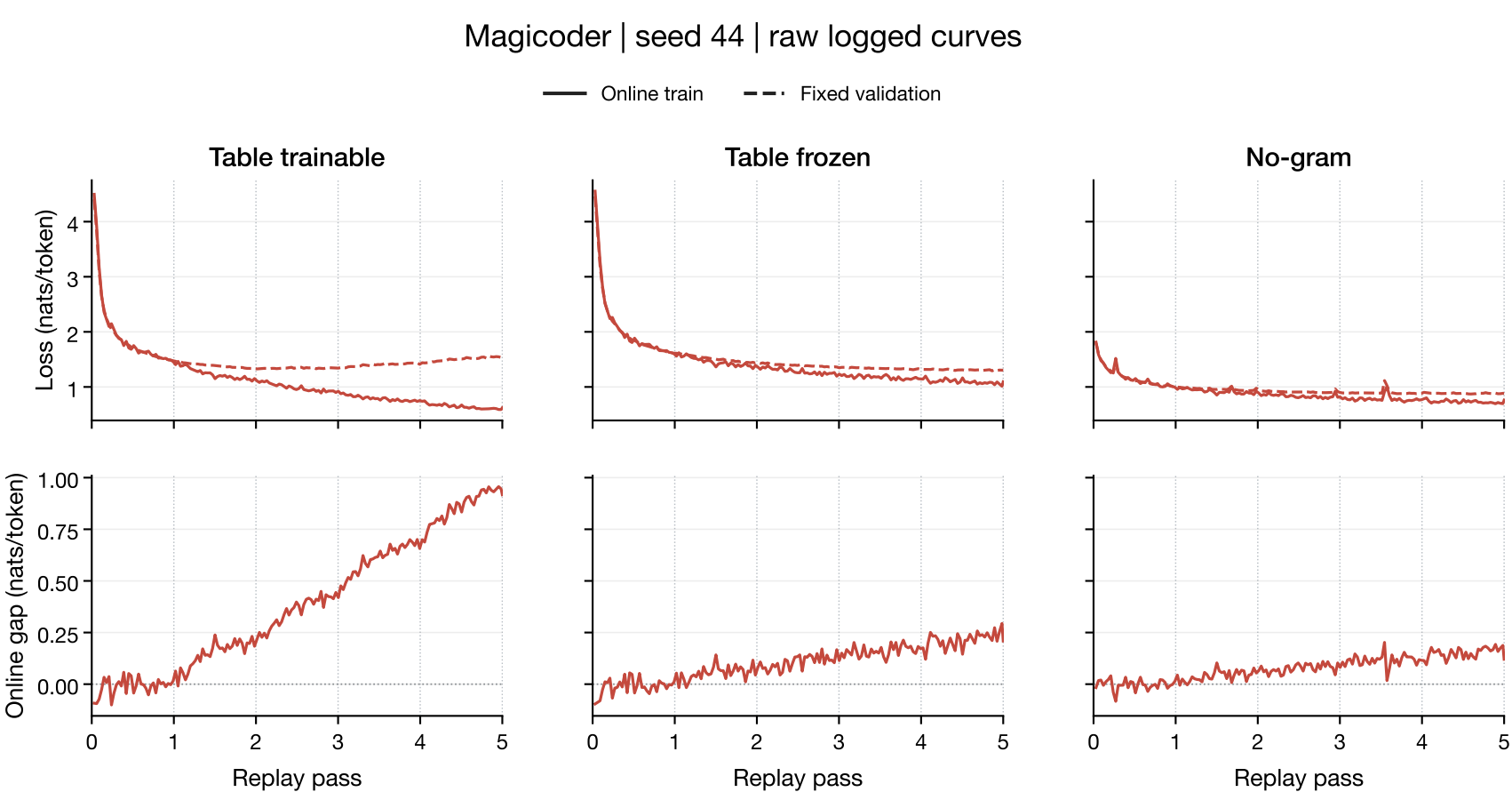

OpenR1 | correct tokenizer | seeds 42, 43, 44
Online train
Fixed validation
Individual seeds
Mean (band: ±1 SD)
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Shared y scales across arms. Gap = fixed validation after update − current-batch train before update.

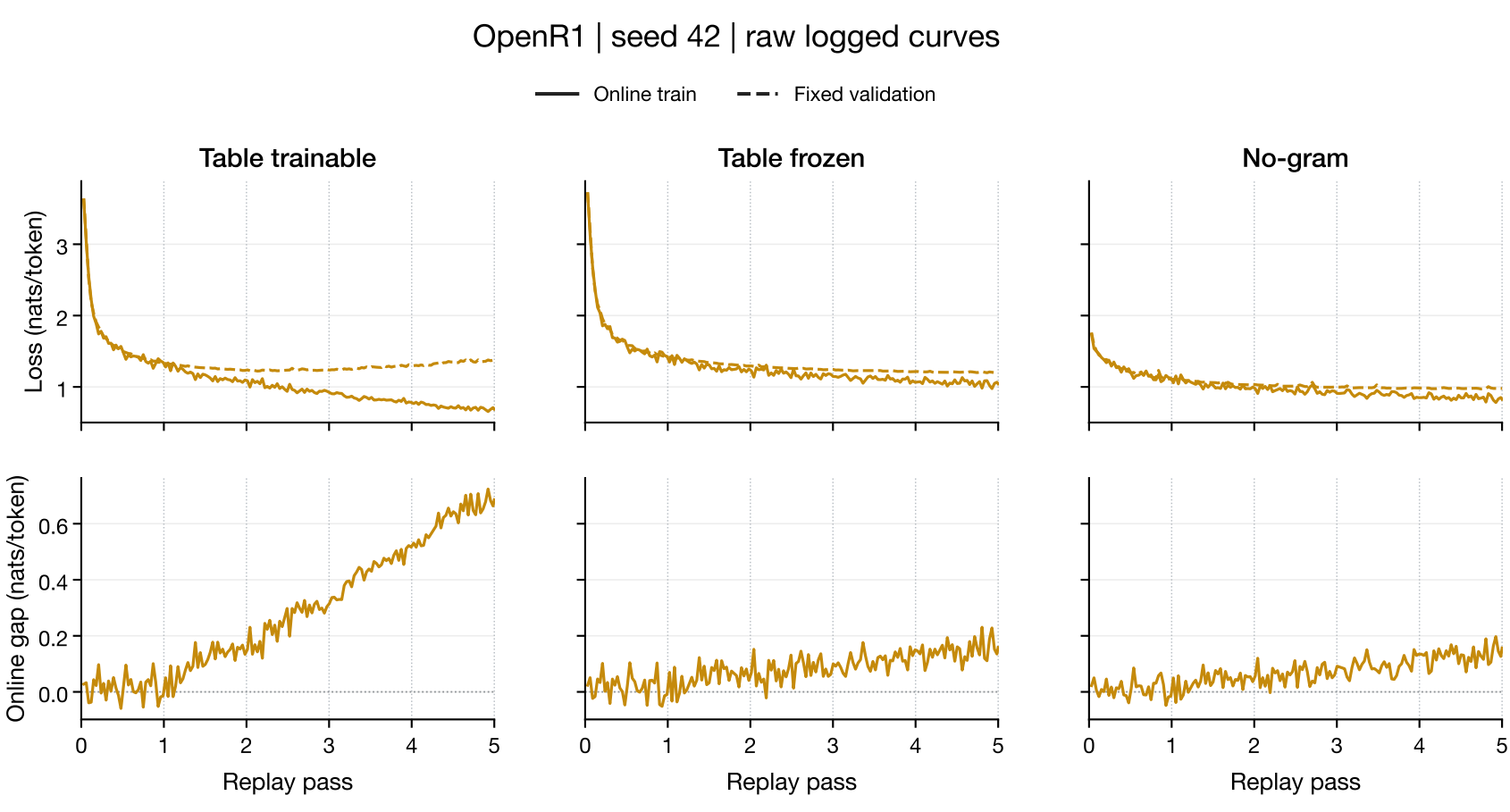

RLVR-MATH (static) | correct tokenizer | seeds 42, 43, 44
Online train
Fixed validation
Individual seeds
Mean (band: ±1 SD)
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Shared y scales across arms. Gap = fixed validation after update − current-batch train before update.

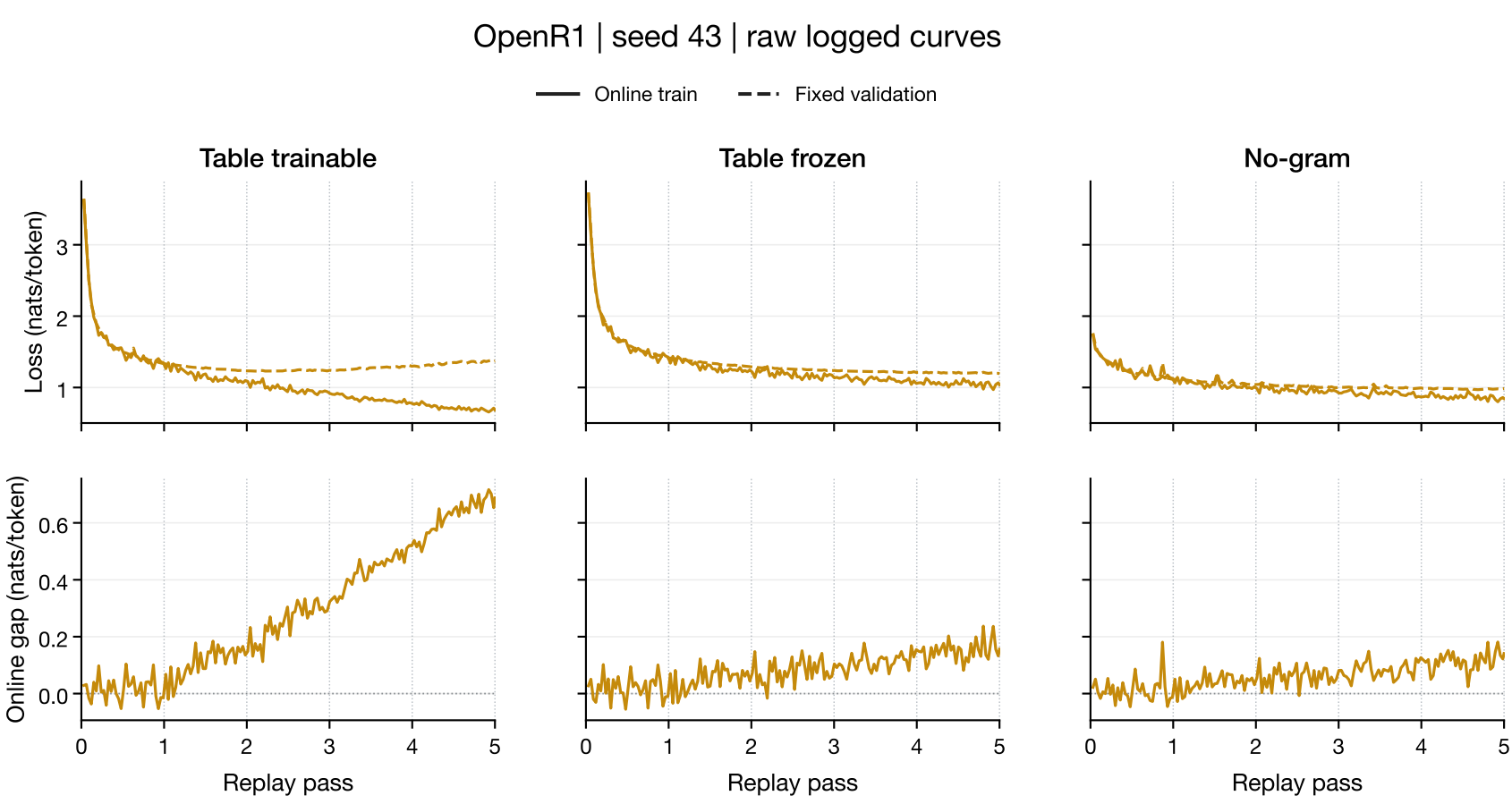

ID | seed 42 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass

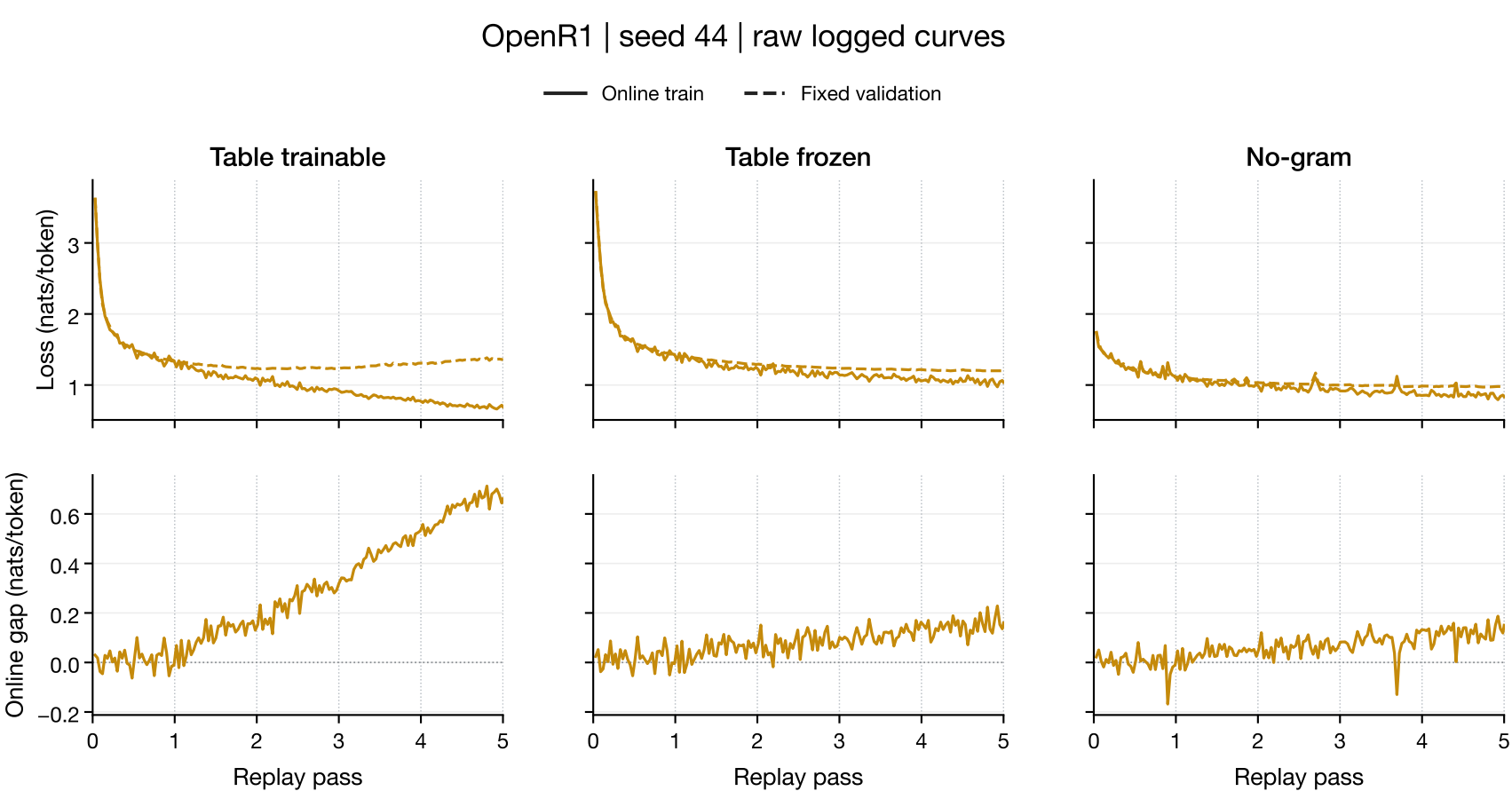

Magicoder | seed 42 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass

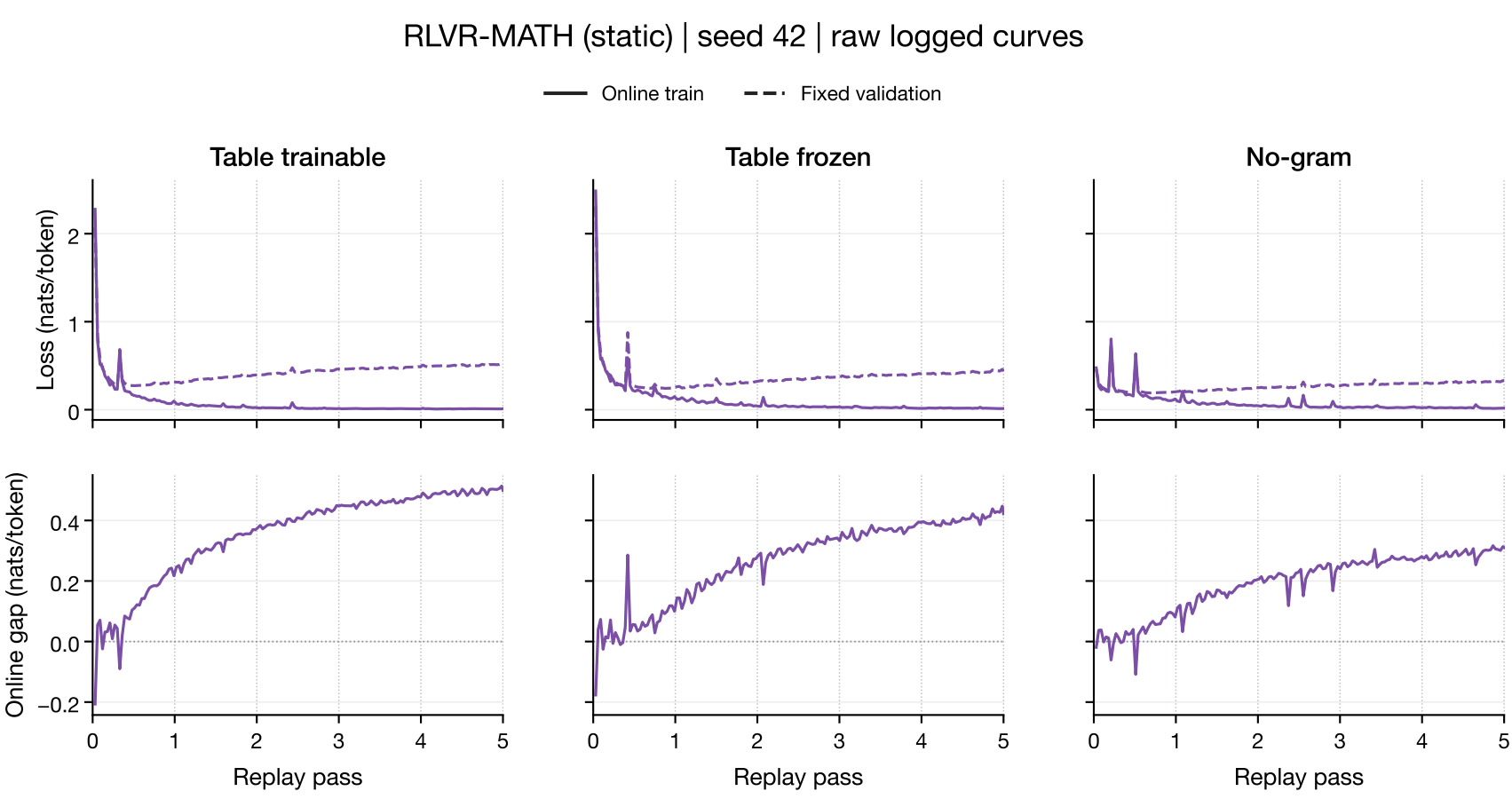

Magicoder | seed 43 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass

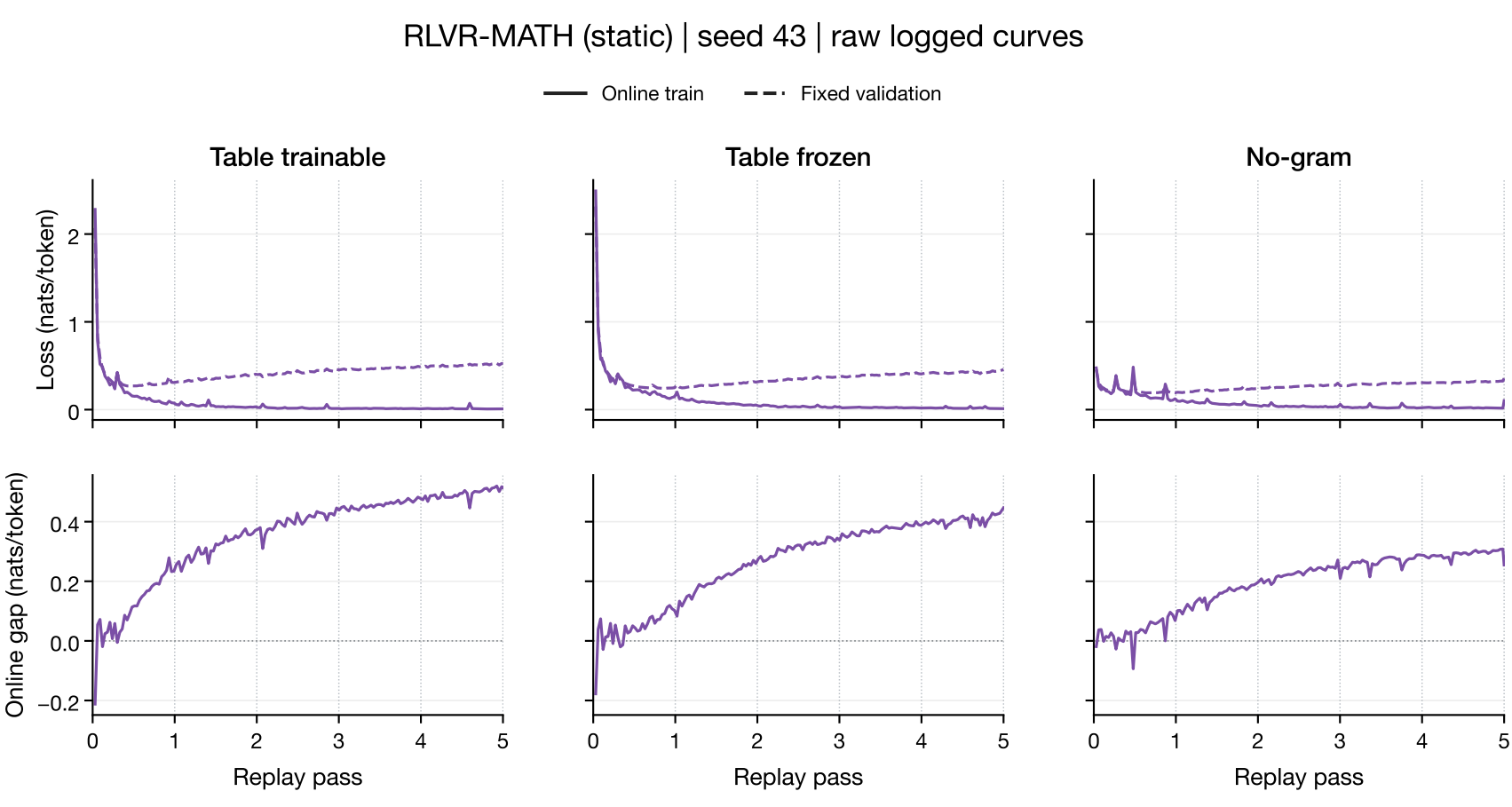

Magicoder | seed 44 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass

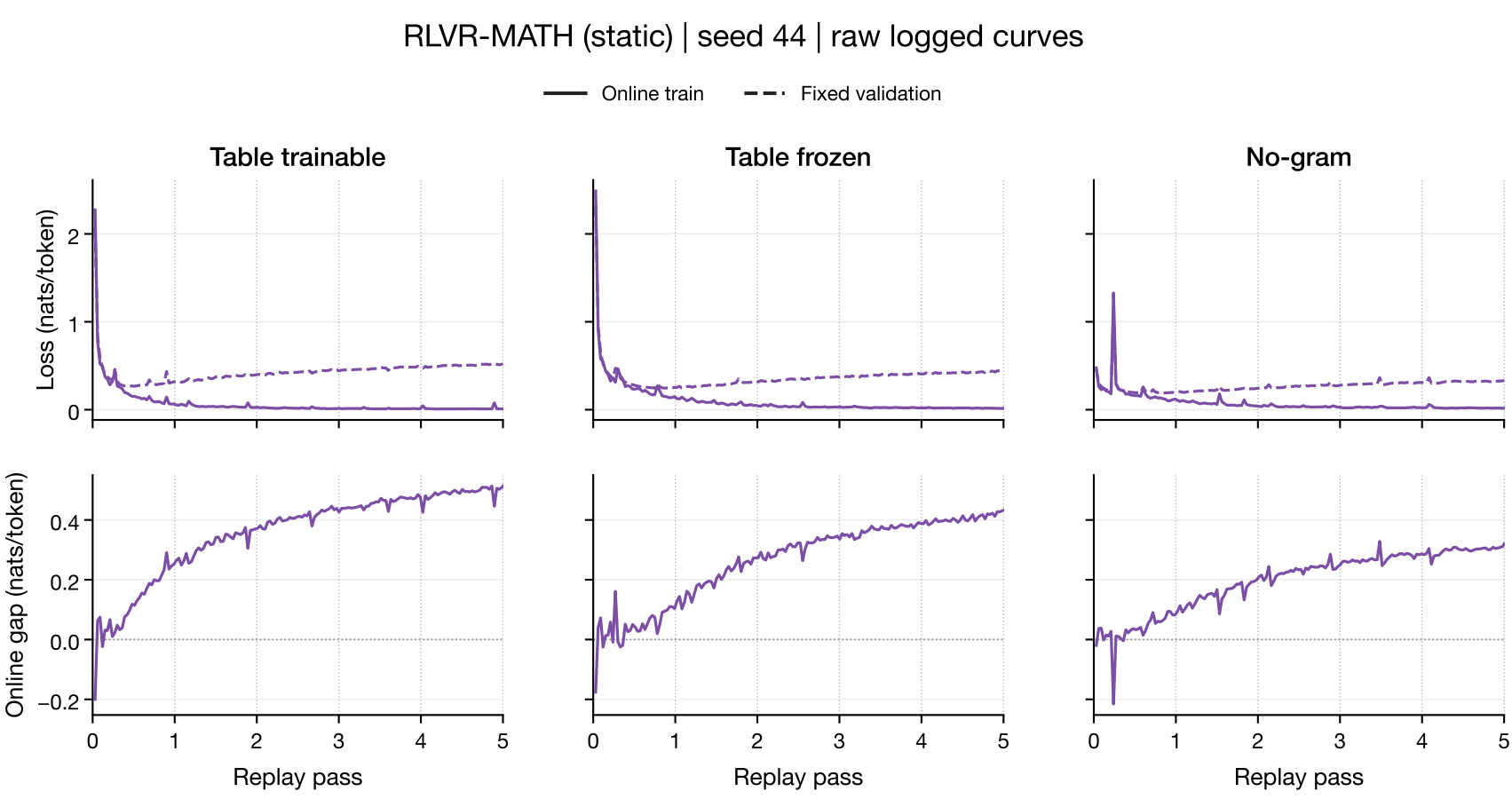

OpenR1 | seed 42 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass

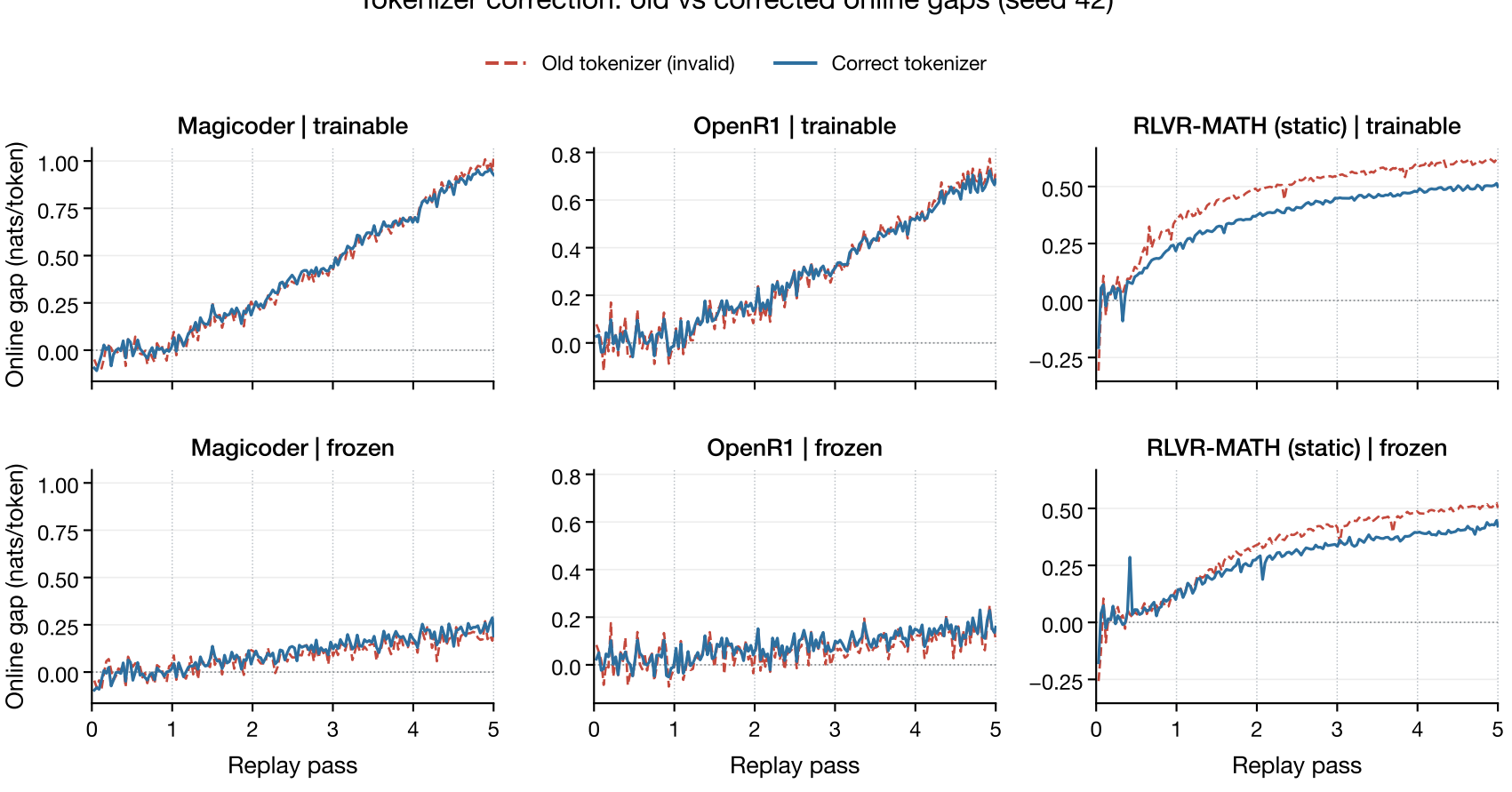

OpenR1 | seed 43 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass

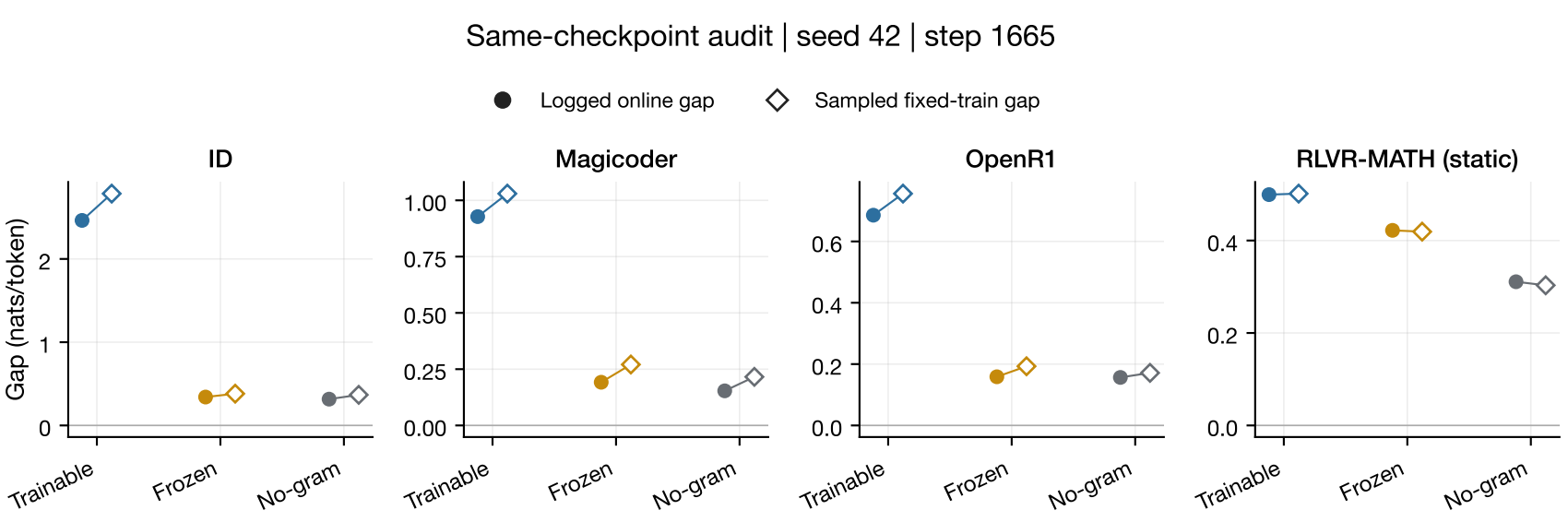

OpenR1 | seed 44 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass


RLVR-MATH (static) | seed 42 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass

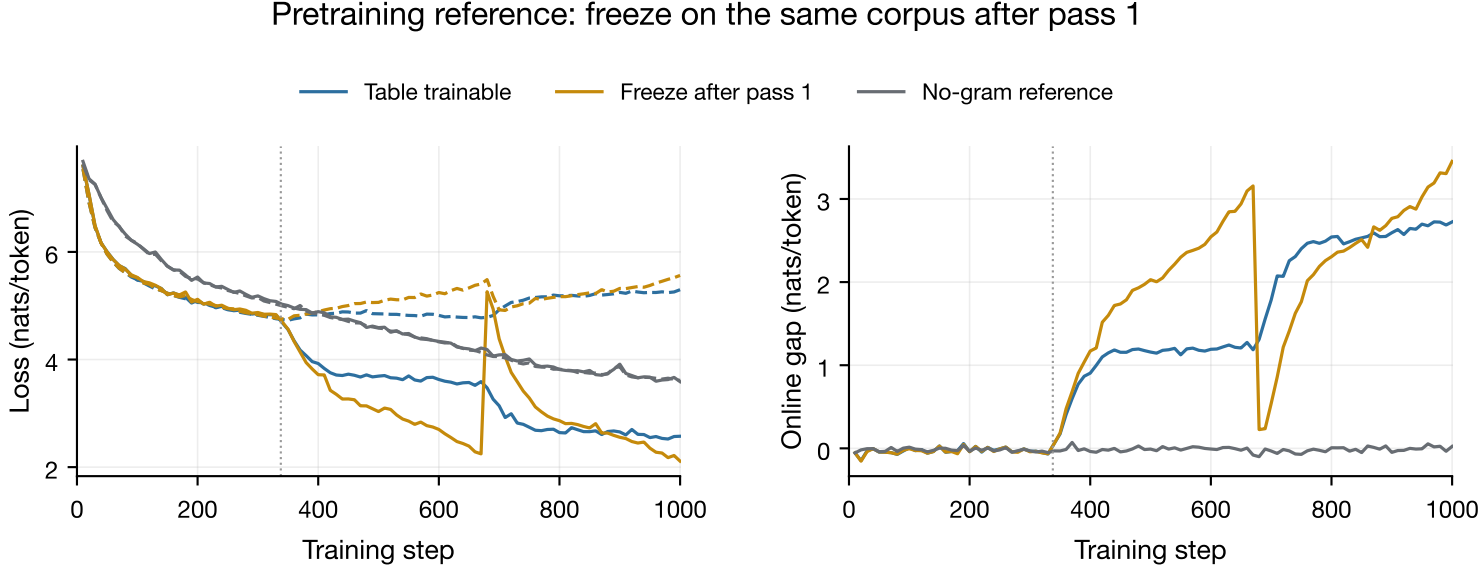

RLVR-MATH (static) | seed 43 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass


RLVR-MATH (static) | seed 44 | raw logged curves
Online train
Fixed validation
Table trainable
Table frozen
No-gram
Loss (nats/token)
Online gap (nats/token)
Replay pass
Replay pass
Replay pass


Tokenizer correction: old vs corrected online gaps (seed 42)
Old tokenizer (invalid)
Correct tokenizer
Magicoder | trainable
OpenR1 | trainable
RLVR-MATH (static) | trainable
Magicoder | frozen
OpenR1 | frozen
RLVR-MATH (static) | frozen
Online gap (nats/token)
Replay pass
Encoding also changes token packing; old/new differences are not a pure matched-text causal effect.



Fixed diagnostic: 33 train batches spanning the pool; same four validation batches. Separate metric.



Seed 42; input bigram+trigram; train solid, fixed validation dashed. Distinct protocol from post-training.